

Representing Rational Numbers

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A **rational number** can be written as a fraction of two integers where the denominator is not zero.

Three Forms

Rational numbers may be represented as **fractions, decimals, or percentages.**

Fraction to Decimal

To change a **fraction** to a **decimal**, divide the numerator by the denominator.

$$\frac{3}{4} = 3 \div 4 = 0.75$$

Decimal to Percentage

To change a **decimal** to a **percentage**, multiply by 100 and attach the percent sign.

$$0.75 \times 100 = 75\%$$

Percentage to Decimal

To change a **percentage** to a **decimal**, divide by 100.

$$65\% \div 100 = 0.65$$

Percentage to Fraction

To change a **percentage** to a **fraction**, write the percentage over 100 and simplify.

$$25\% = \frac{25}{100} = \frac{1}{4}$$

Equivalent Forms

Equivalent fractions, decimals, and percentages represent the **same value** in different forms.

$$\frac{3}{4} = 0.75 = 75\%$$

Examples

Example 1

Express $\frac{3}{4}$ as a decimal and a percentage.

$$\frac{3}{4} = 0.75 = 75\%$$

Example 2

Express 0.45 as a fraction in simplest form and as a percentage.

$$0.45 = \frac{45}{100} = \frac{9}{20} = 45\%$$

Example 3

Express 65% as a decimal and as a fraction in simplest form.

$$65\% = 0.65 = \frac{65}{100} = \frac{13}{20}$$

Practice Exercises

Practice 1

Express $\frac{3}{4}$ as a decimal and a percentage.

Practice 2

Express $\frac{7}{20}$ as a decimal and a percentage.

Practice 3

Express 0.45 as a fraction in simplest form and as a percentage.

Practice 4

Express 0.125 as a fraction in simplest form and as a percentage.

Practice 5

Express 65% as a decimal and as a fraction in simplest form.

Practice 6

Express 12% as a decimal and as a fraction in simplest form.

Practice 7

A notebook is sold at a 25% discount. Express the discount as a decimal and as a fraction.

Activities

Activity 1

Express $\frac{5}{8}$ as a decimal and a percentage.

Activity 2

Express $\frac{9}{25}$ as a decimal and a percentage.

Activity 3

Express 0.35 as a fraction in simplest form and as a percentage.

Activity 4

Express 0.625 as a fraction in simplest form and as a percentage.

Activity 5

Express 72% as a decimal and as a fraction in simplest form.

Activity 6

Express 18% as a decimal and as a fraction in simplest form.

Activity 7

A store offers a 40% discount on a bag. Express the discount as a decimal and as a fraction.

Thank you!